

Tusher Karmakar

Ph.D. Student, Electrical and Computer Engineering, Iowa State University

✉ karmakartusher24@gmail.com

🌐 github.com/TusherKarmakar

Research Interests: Matrix Factorization, Machine Learning, Signal Processing, Data Privacy

Education

Iowa State University (ISU), Ames, Iowa, USA

Aug 2025 – Present

Ph.D. in Electrical and Computer Engineering (ECpE)

Advisor: Prof. Namrata Vaswani

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

2019 – 2024

B.Sc. in Electrical and Electronic Engineering

Major: Communication and Signal Processing; CGPA: 3.63/4.00

Undergraduate Thesis: *Privacy-Preserving Non-negative Matrix Factorization for Decentralized Data Using Correlated Noise*

Supervisor: Prof. Hafiz Imtiaz

Experience

Graduate Research Assistant, Iowa State University

May 2026 – Present

Department of Electrical and Computer Engineering (ECpE)

Advisor: Prof. Namrata Vaswani

- Working on Robust Principal Component Analysis (RPCA) and low-rank matrix factorization methods for video background subtraction and experimental evaluation.

Graduate Teaching Assistant, Iowa State University

Aug 2025 – May 2026

Department of Electrical and Computer Engineering (ECpE)

- Courses Assisted: EE 4420 (Introduction to Circuits and Systems), EE 2010 (Electric Circuits), EE 4480 (Introduction to AC Circuits and Motors)
- Responsibilities: grading, office-hour tutoring, laboratory instruction, and review-session support

Tutor (Mathematics Educator), Cymath

Aug 2025 – Present

Undergraduate Research Assistant, BUET

Jan 2023 – Apr 2024

Department of Electrical and Electronic Engineering

Supervisor: Prof. Hafiz Imtiaz

- Worked on privacy-preserving non-negative matrix factorization for decentralized data systems, leading to a journal publication in *Signal, Image and Video Processing*.

Selected Publications

1. Hafiz Imtiaz, **Tusher Karmakar**, Protoye Kumar Mohanata, “Privacy-Preserving Non-negative Matrix Factorization for Decentralized Data Using Correlated Noise,” *Signal, Image and Video Processing*, 2025. doi: 10.1007/s11760-025-03887-1
2. Sandipa Chowdhury, Mohtasim Billah, Sudipto Pramanik, Shaikh Anowarul Fattah, **Tusher Karmakar**, “Bidirectional Cross-Dataset Transfer Learning for Human Activity Recognition with Dataset-Specific Adapters and EWC,” *11th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering*, 2025. doi: 10.1109/WIECON-ECE69386.2025.11526593

Awards and Honors

- **Best Teaching Assistant Award**, Department of Electrical and Computer Engineering, Iowa State University, 2026

Technical Skills

Programming: Python, MATLAB, C/C++

Machine Learning: PyTorch, TensorFlow

Hardware and Simulation: Verilog, FPGA, HFSS, PSPICE, Proteus, Quartus
Documentation: L^AT_EX